

Option C (Algorithm and Programming)

Booking System

In today's busy world, booking systems are widely used to reserve services and facilities. A booking system enables users to browse available services or resources, check availability, make reservations, and manage their bookings efficiently.

Students should implement a booking system for **one specific type of usage**. Examples include:

- Cinema Ticket Booking System
- Concert / Event Ticket Booking System
- Restaurant Reservation System
- Hospital / Clinic Appointment System
- School Room / Facility Booking System
- Any other suitable booking system (e.g. sports venue, library seat, hotel room, etc.)

Task 1 (Design & Implementation)

Write a program to implement a Booking System for the chosen type of usage. You are advised to include at least three distinct features or modules. The program should handle a reasonable amount of data (e.g. at least 10 - 15 events, showtimes, time slots or resources). Complete the following tasks:

- (a) Select suitable data types and data structures for your system elements.
- (b) Implement functions for searching and displaying available events or services.
- (c) Design and implement the booking, seat/time slot selection, and payment simulation process.
- (d) Using stepwise refinement, describe the algorithms for the main menu loop and key features.
- (e) Use a flowchart to describe the algorithms in (d).

You may want to consider some of the following key factors when designing the program:

- selection of data types and data structures
- variable/constant declaration and initialisation
- modular approach
- scope of variables and parameters passing
- interface of the program
- data collection, input and validation
- data processing
- sorting and searching algorithms
- program output
- error handling
- reusability and portability

Create a presentation and/or documents to briefly describe the components involved in designing the program.

Task 2 (Testing & Evaluation)

Referring to the program (*Alternative: Using a program stipulated by your teacher*), complete the following tasks.

Conduct a test of the program. Collect and record the feedback and results of the test.

- Either (i) make one major change in the program and illustrate the corresponding improvement,
Or (ii) describe how the scope of the program could be extended.

Create a presentation and/or documents to illustrate the development of the program. You may want to consider some of the following:

- pros and cons of the program design
- test data and test cases
- unit test
- system test
- user acceptance test
- debugging
- algorithm optimisation